

Universidade do Minho
Escola de Engenharia
Departamento de Electrónica Industrial

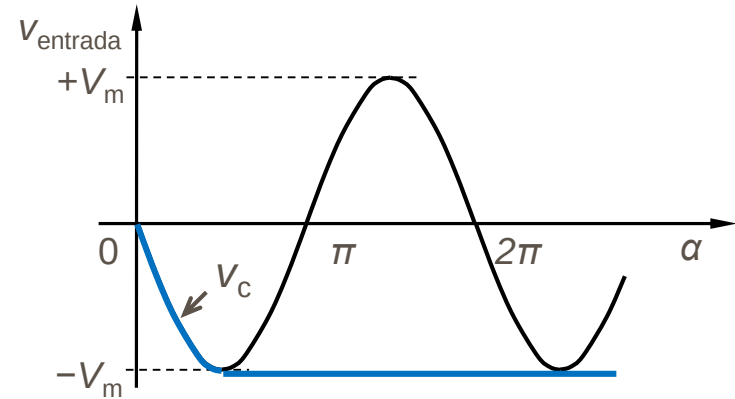
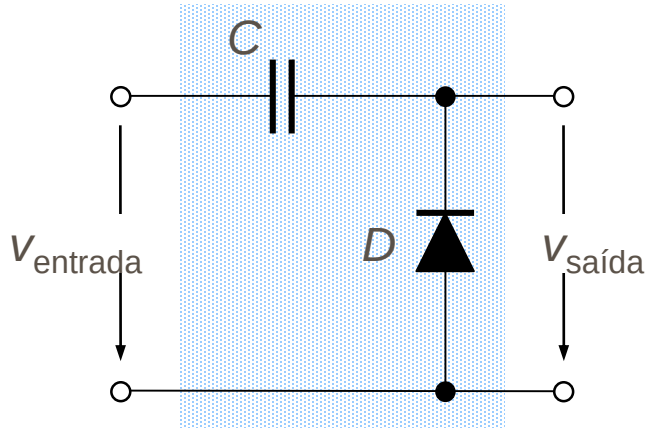
Engenharia de Telecomunicações e Informática

Eletrónica Geral

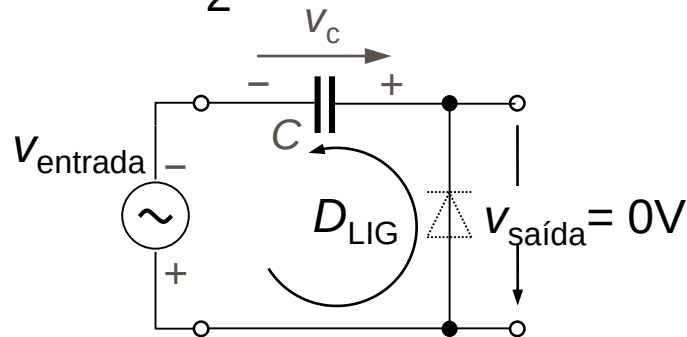
Departamento de Eletrónica Industrial

Díodos – Aplicações Básicas

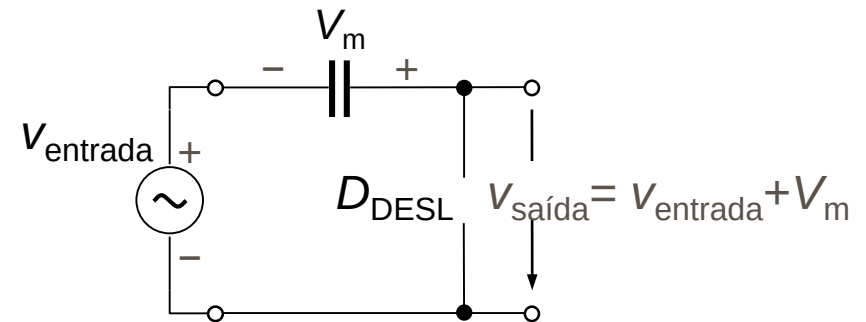
■ Circuitos Fixadores (ou *Clamping*)



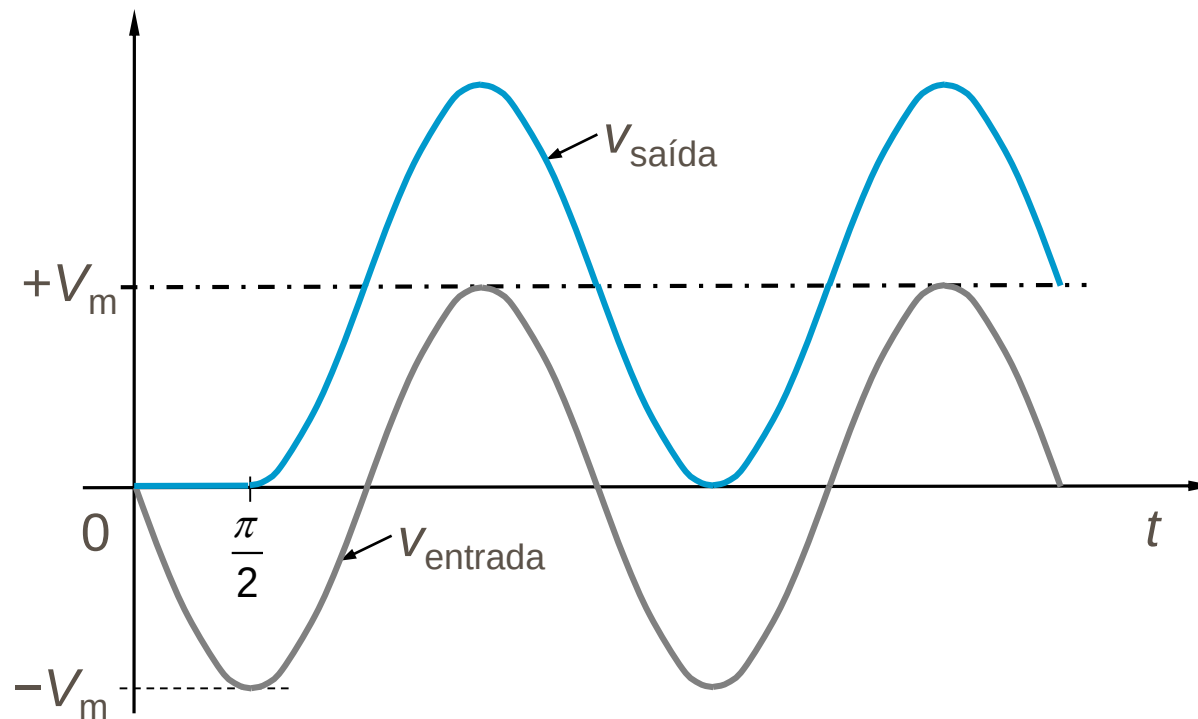
$\alpha \leq \frac{\pi}{2}$ (diodo conduz)



$\alpha > \frac{\pi}{2}$ (diodo não conduz)



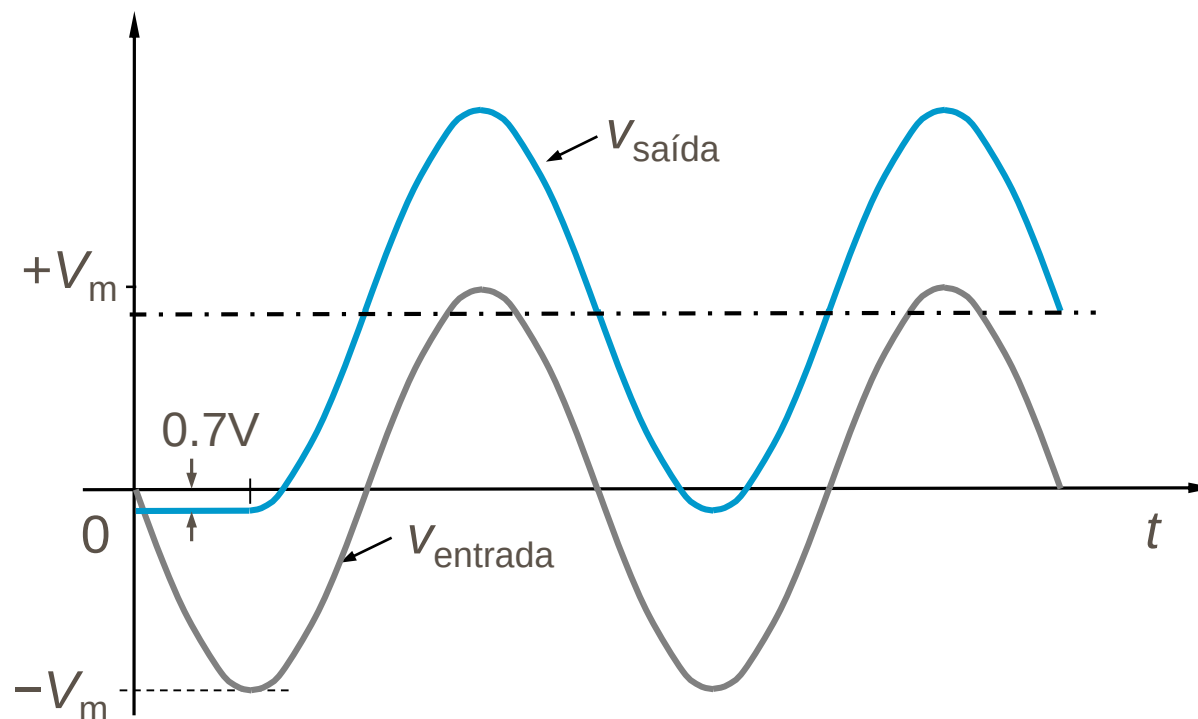
■ Circuitos Fixadores



Fixador – formas de onda admitindo diodo ideal

$$V_{\text{saída}} = V_{\text{entrada}} + V_m = -V_m \sin \omega t + V_m$$

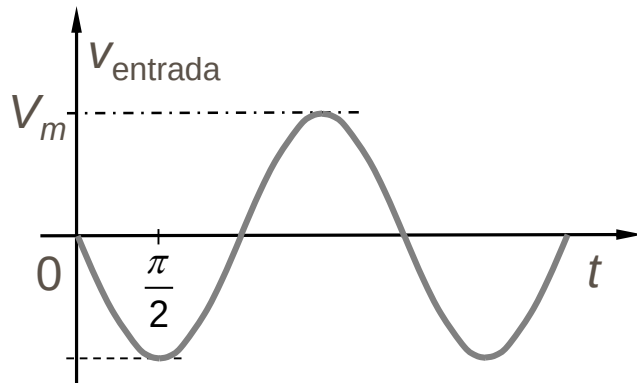
■ Circuitos Fixadores



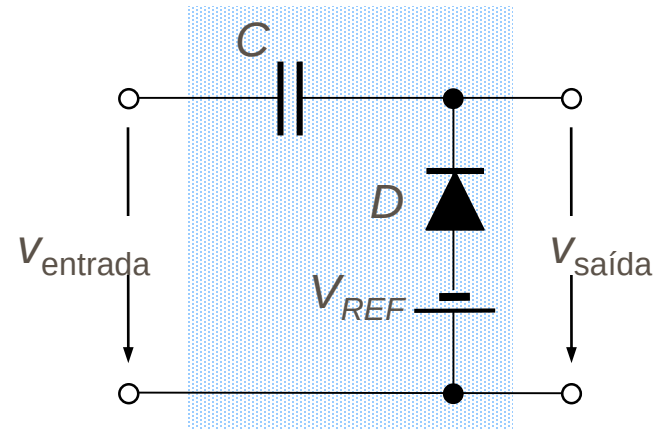
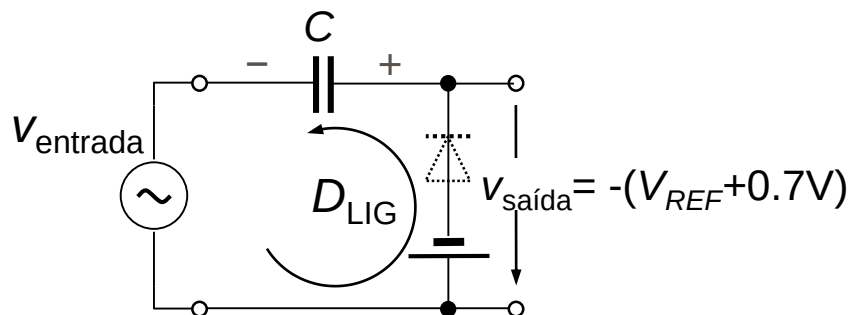
Fixador – formas de onda no caso do díodo real

Díodos – Aplicações Básicas

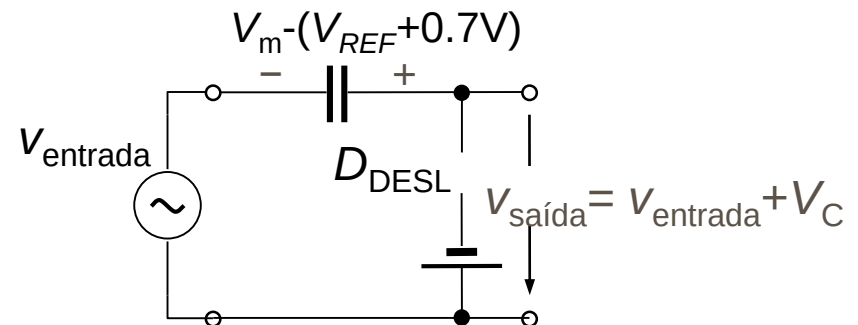
■ Circuitos Fixadores (caso geral)



$$\alpha \leq \frac{\pi}{2} \text{ (diodo conduz)}$$

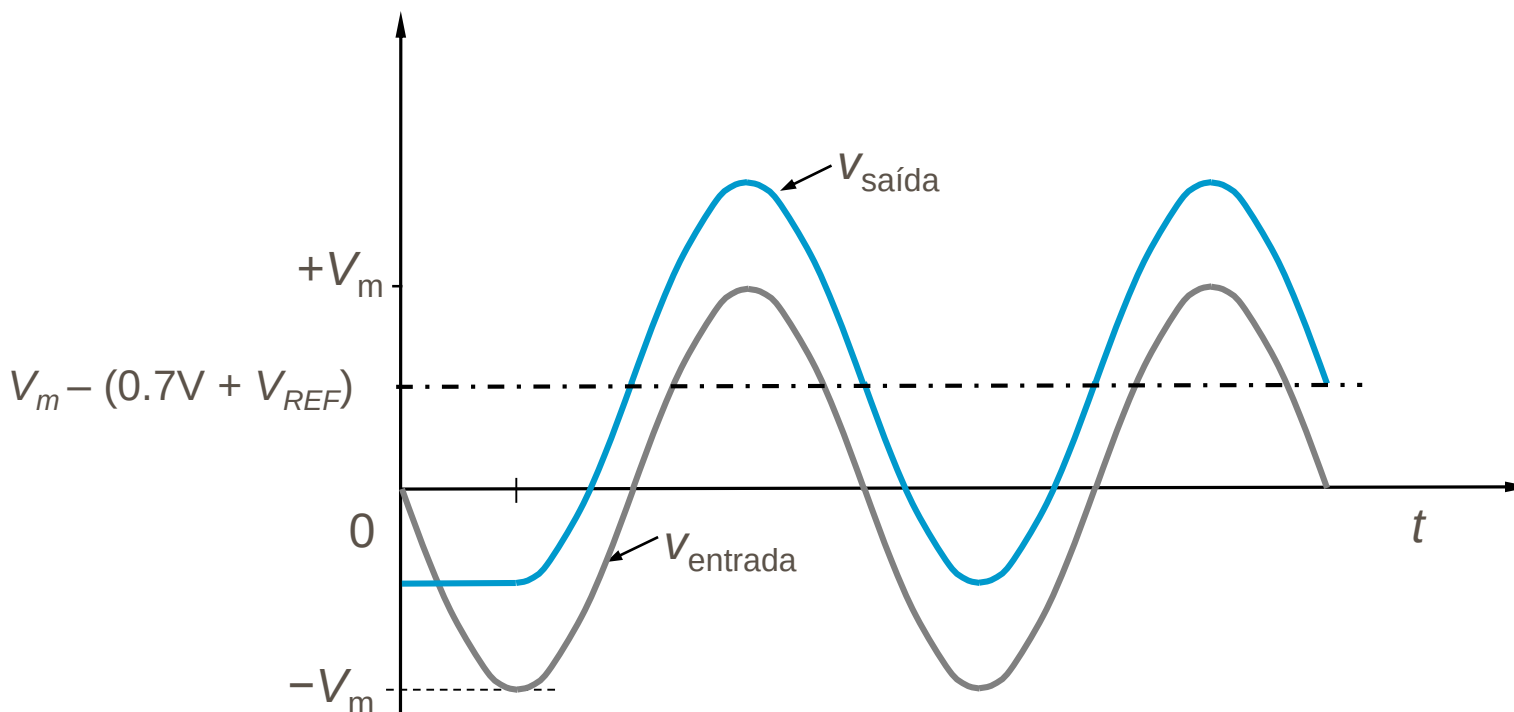


$$\alpha > \frac{\pi}{2} \text{ (diodo não conduz)}$$



Díodos – Aplicações Básicas

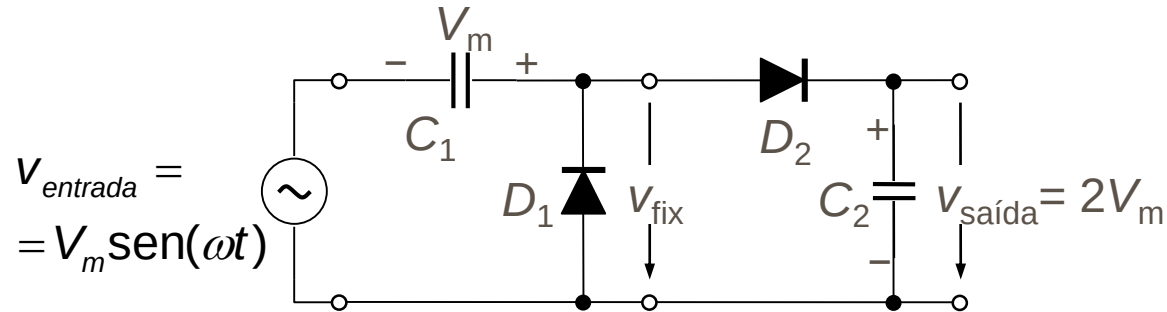
■ Circuitos Fixadores (caso geral)



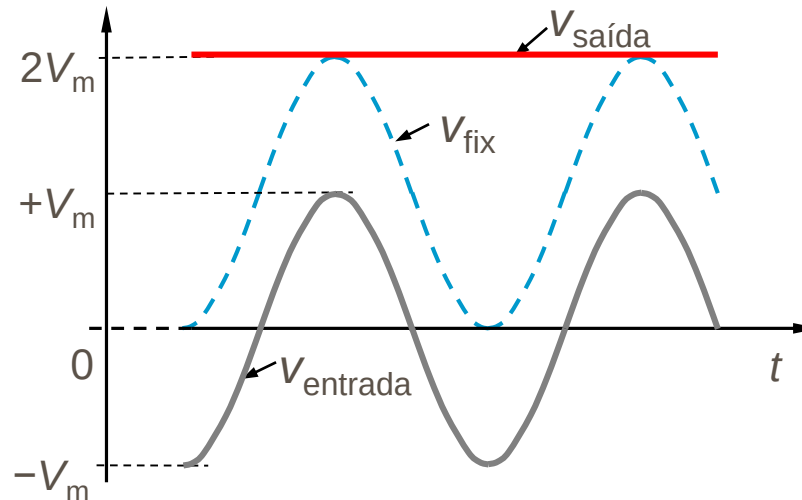
Fixador – formas de onda no caso do díodo real

Díodos – Aplicações Básicas

■ Detector Pico-a-Pico (ou duplicador de tensão)

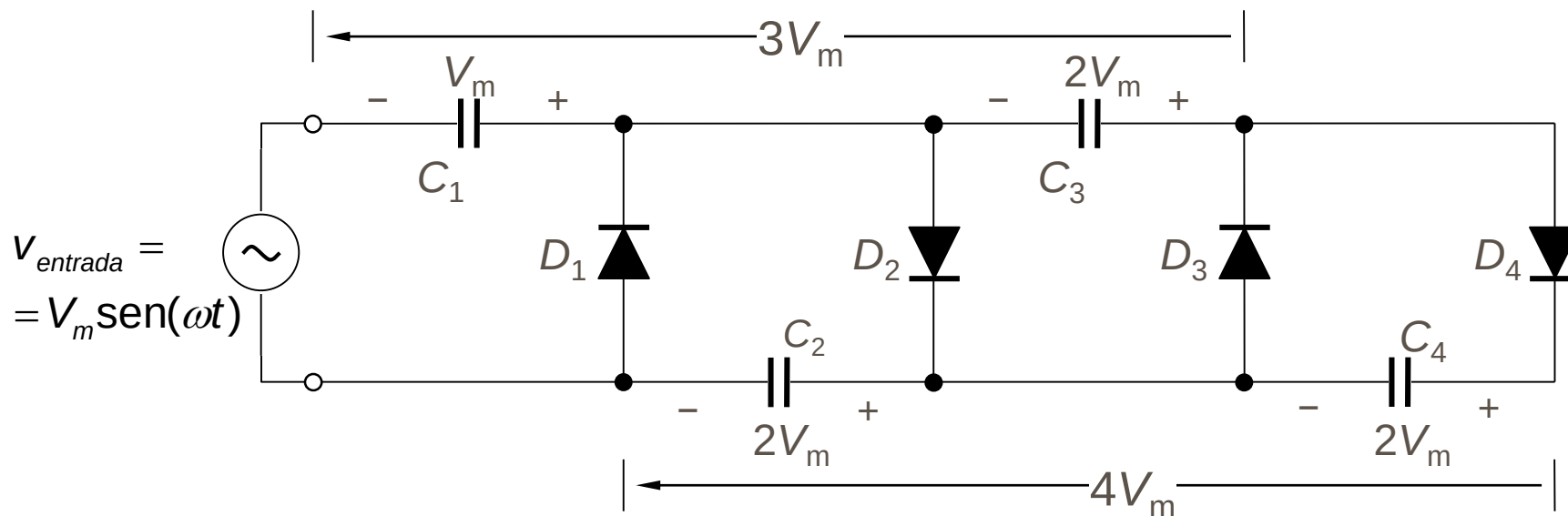


Duplicador – formas de onda admitindo díodo ideal (reg. permanente)



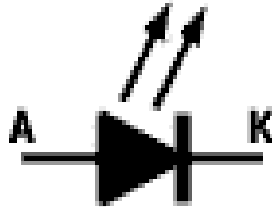
Díodos – Aplicações Básicas

■ Multiplicador de tensão

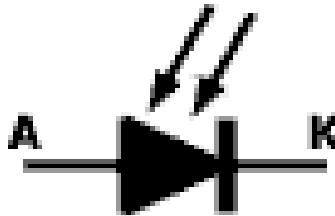


Díodos – Outras Aplicações

- **Díodo emissor de luz (LED- Light Emitting Diode)**
 - Quando percorrido por uma corrente liberta fotões
 - $V_{LED} \approx 1,2 - 1,6V$
 - $i_{LED} \approx 8 - 15 \text{ mA}$

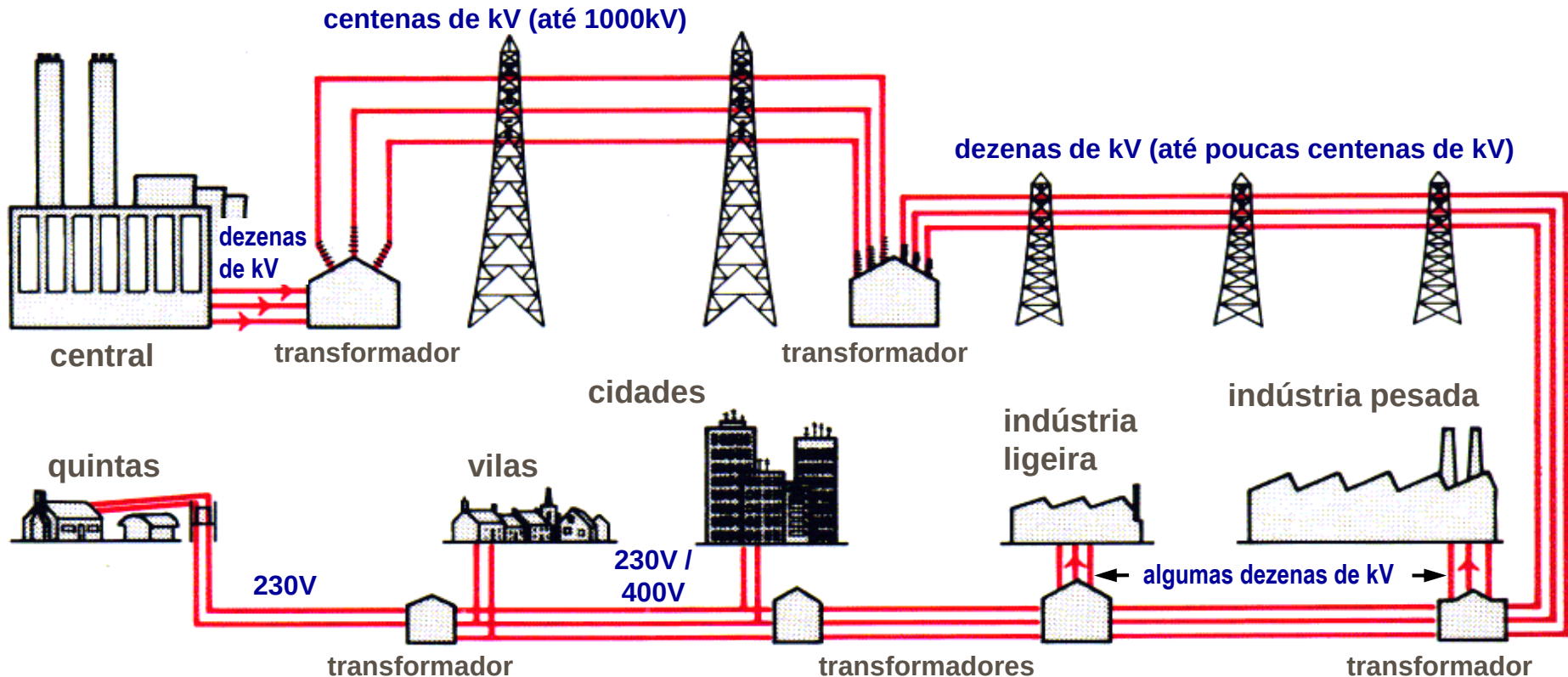


- **Fotodíodos**
 - Quando atingidos por fotões conduzem reversamente polarizados



Fontes de Alimentação

■ Produção Transporte e Distribuição de Energia Eléctrica



Esquema Tradicional de Produção, Transporte e Distribuição de Energia (PTDE)



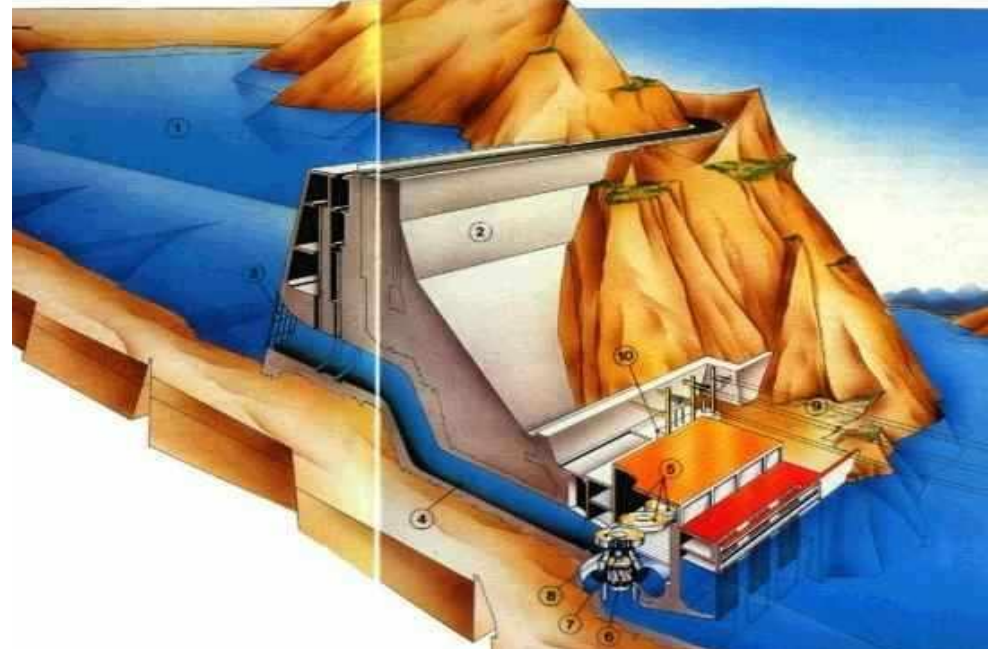
Central Térmica



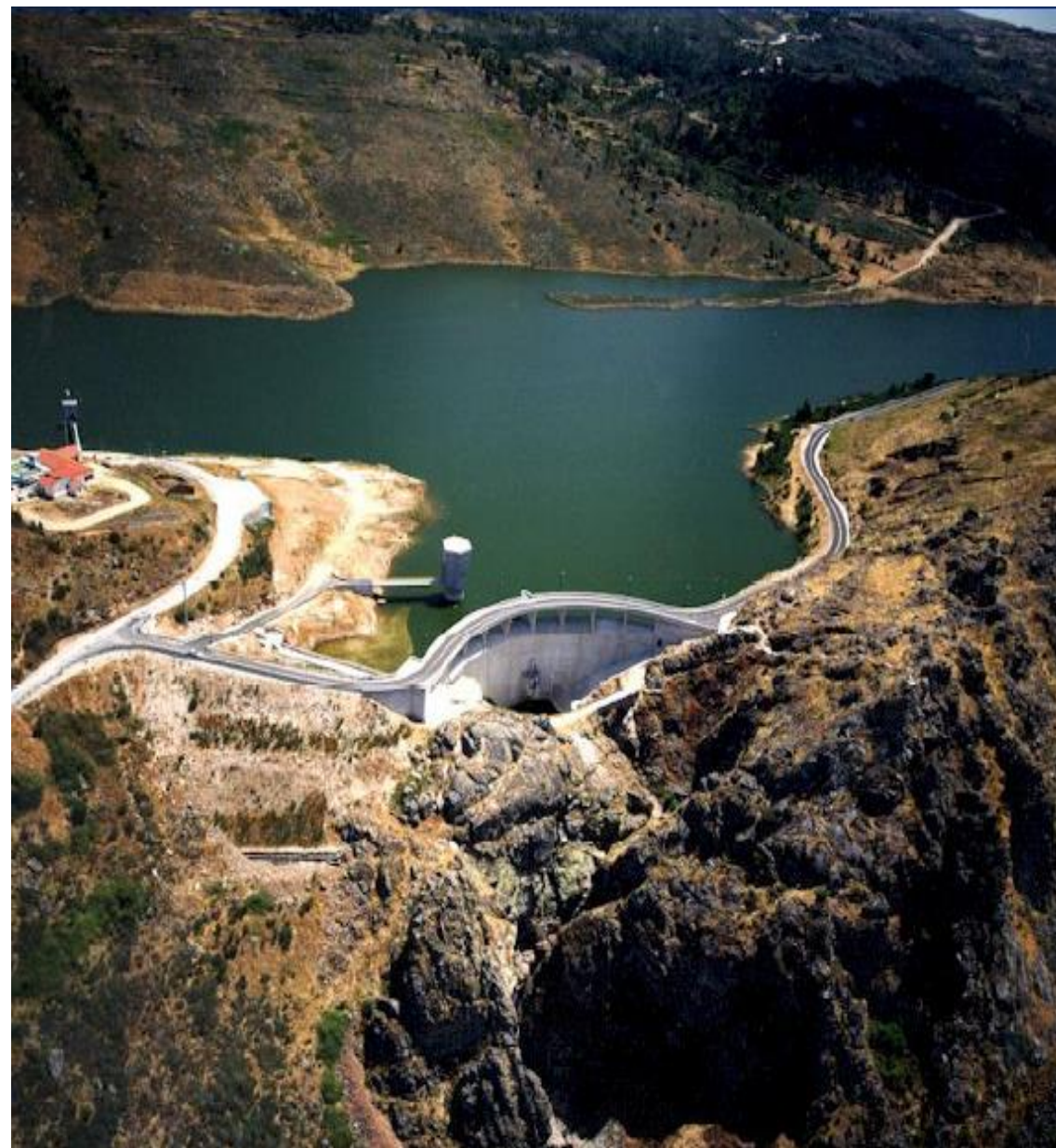
Central Hidroelétrica



Júlio Martins, Central Atómica



Universidade do Minho, Eletrónica Geral (LETI)

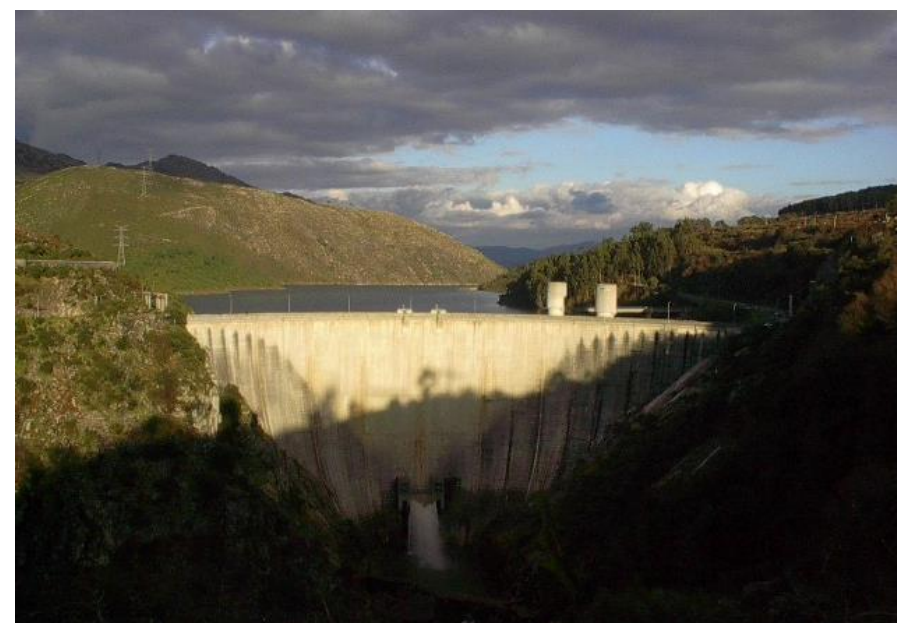


Albufeira do Caldeirão

Júlio Martins,



Aguieira



Alto Lindoso

Universidade do Minho, Eletrónica Geral (LETI)

Transformador de
alta tensão

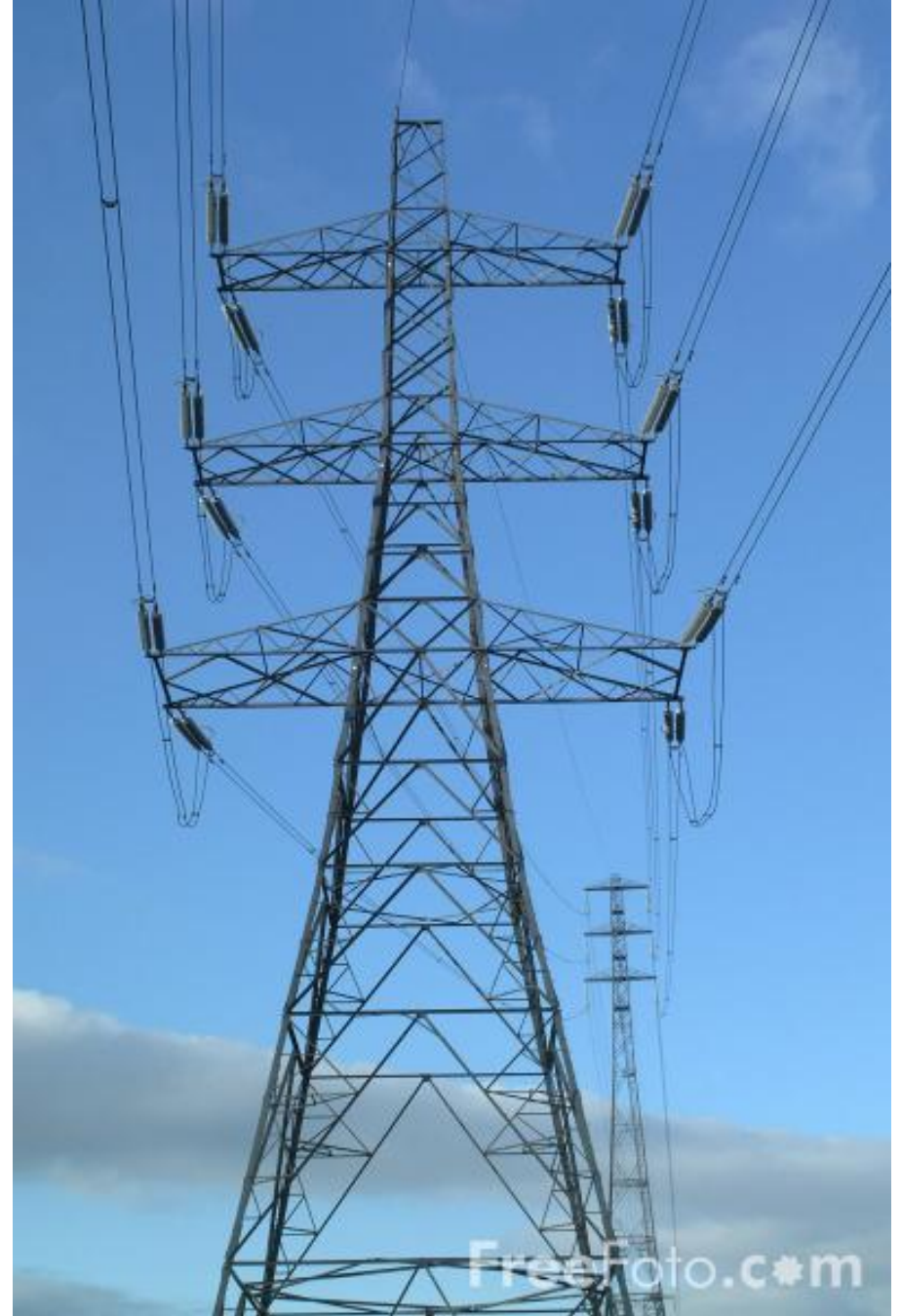


Posto de transformação

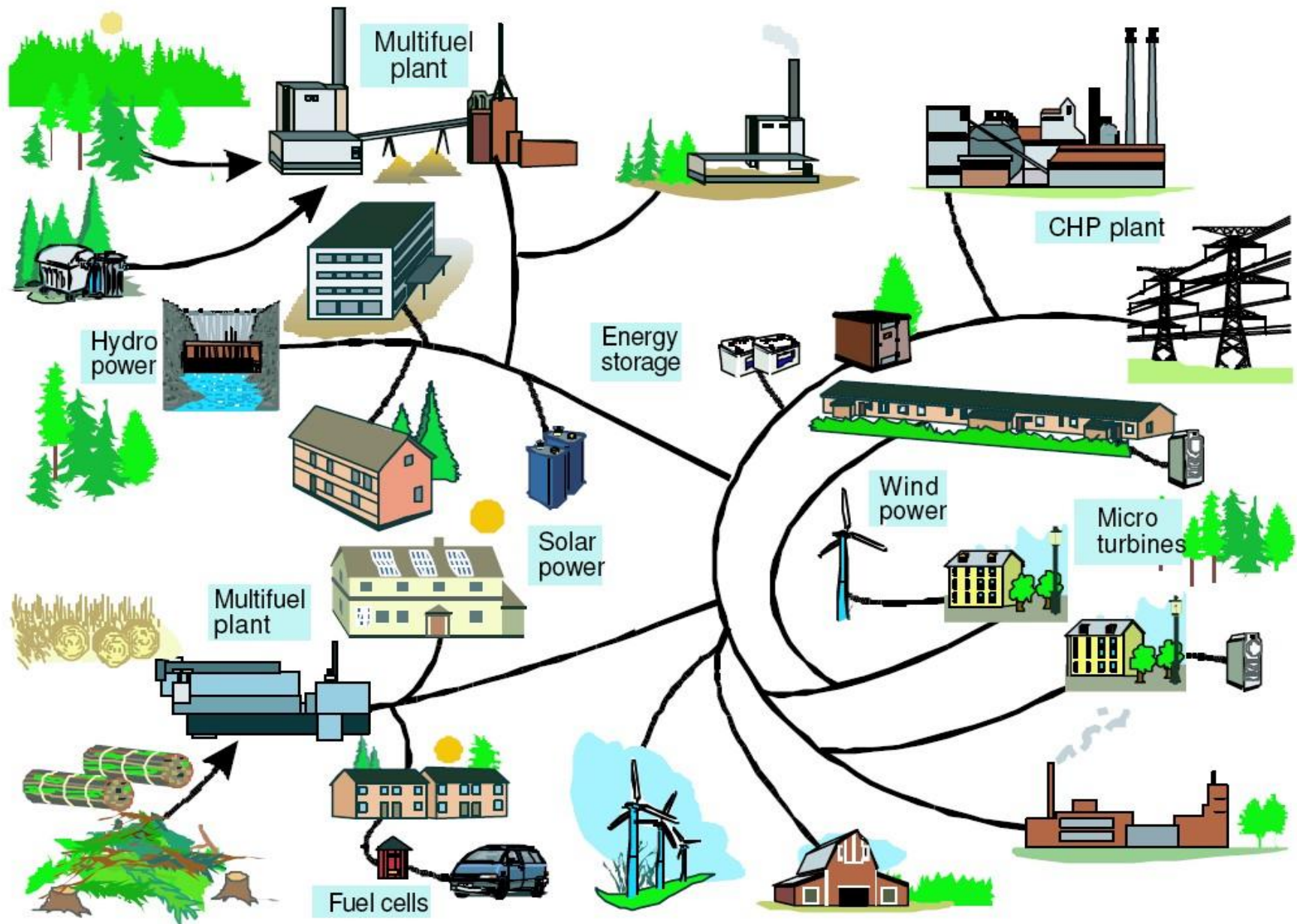


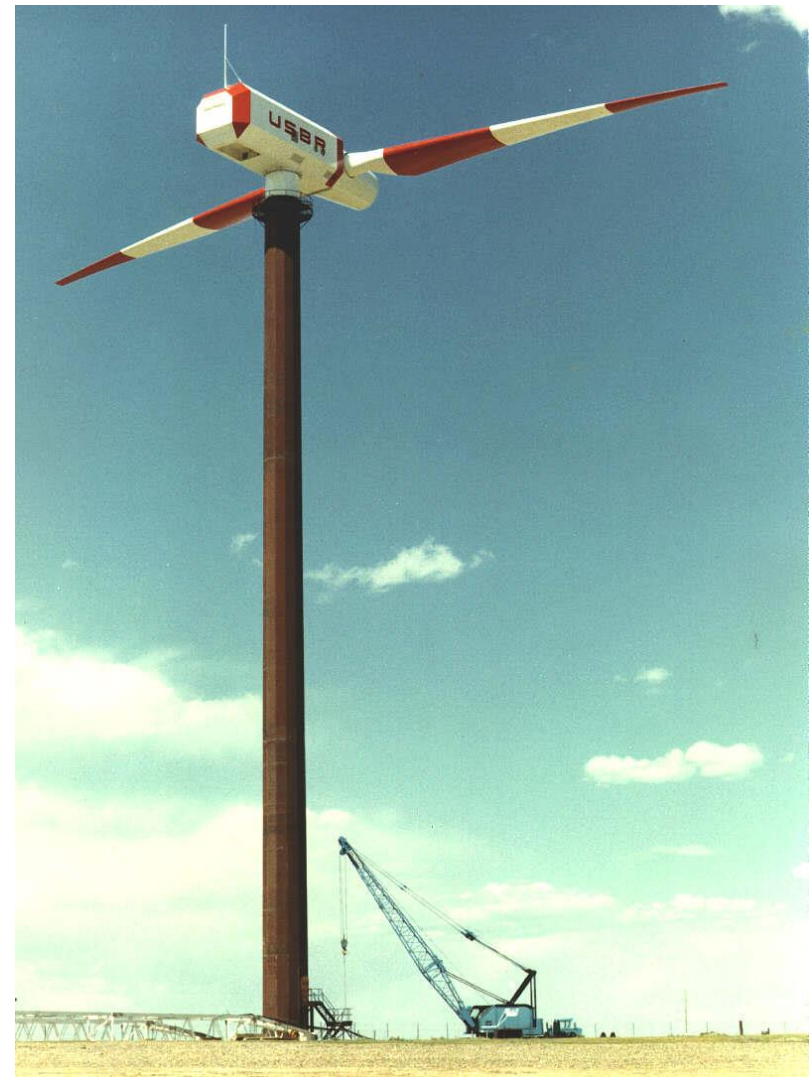


Júlio Martins,



Universidade do Minho, Eletrónica Geral (LETI)



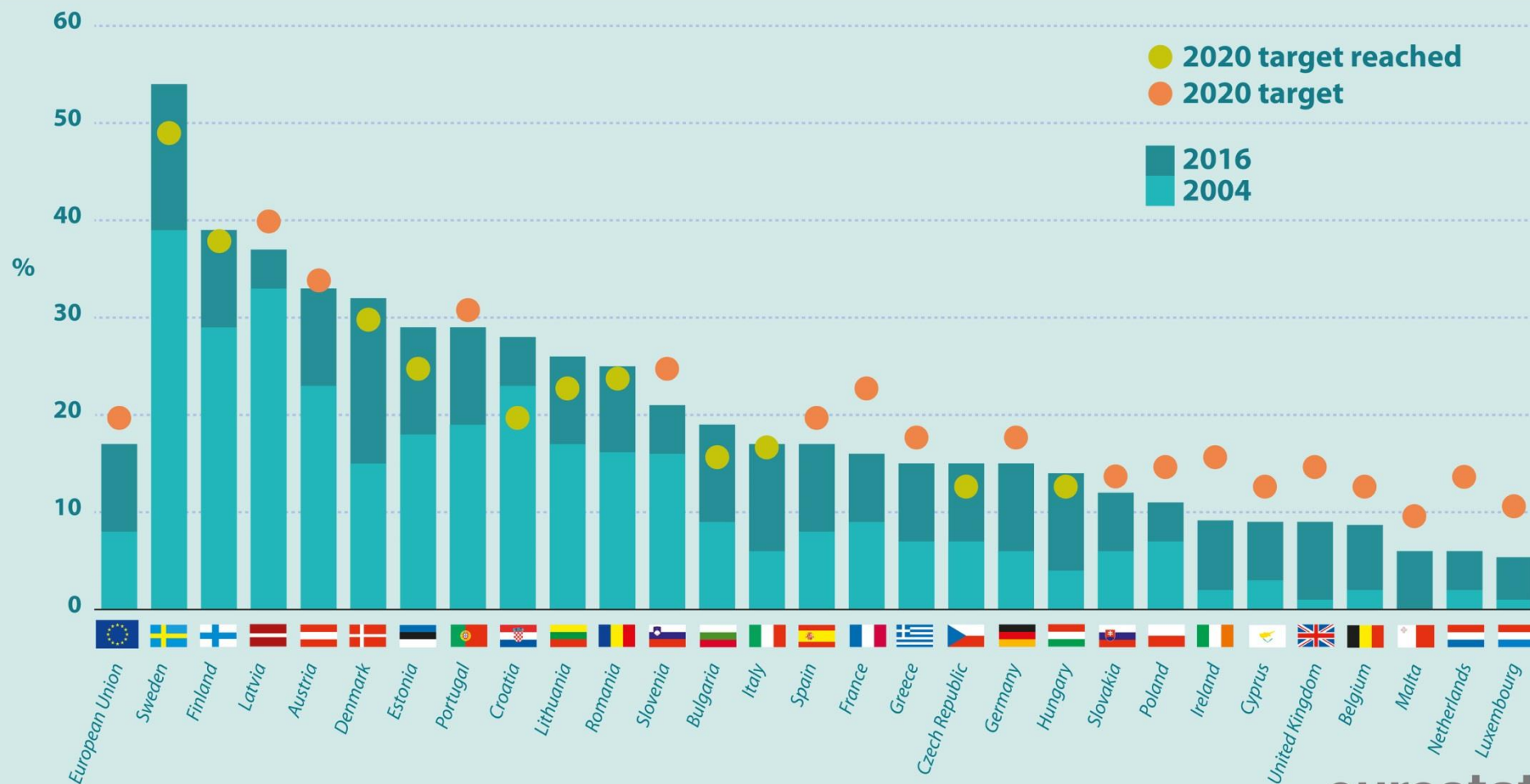


1,18 MWp
10.080 módulos FV

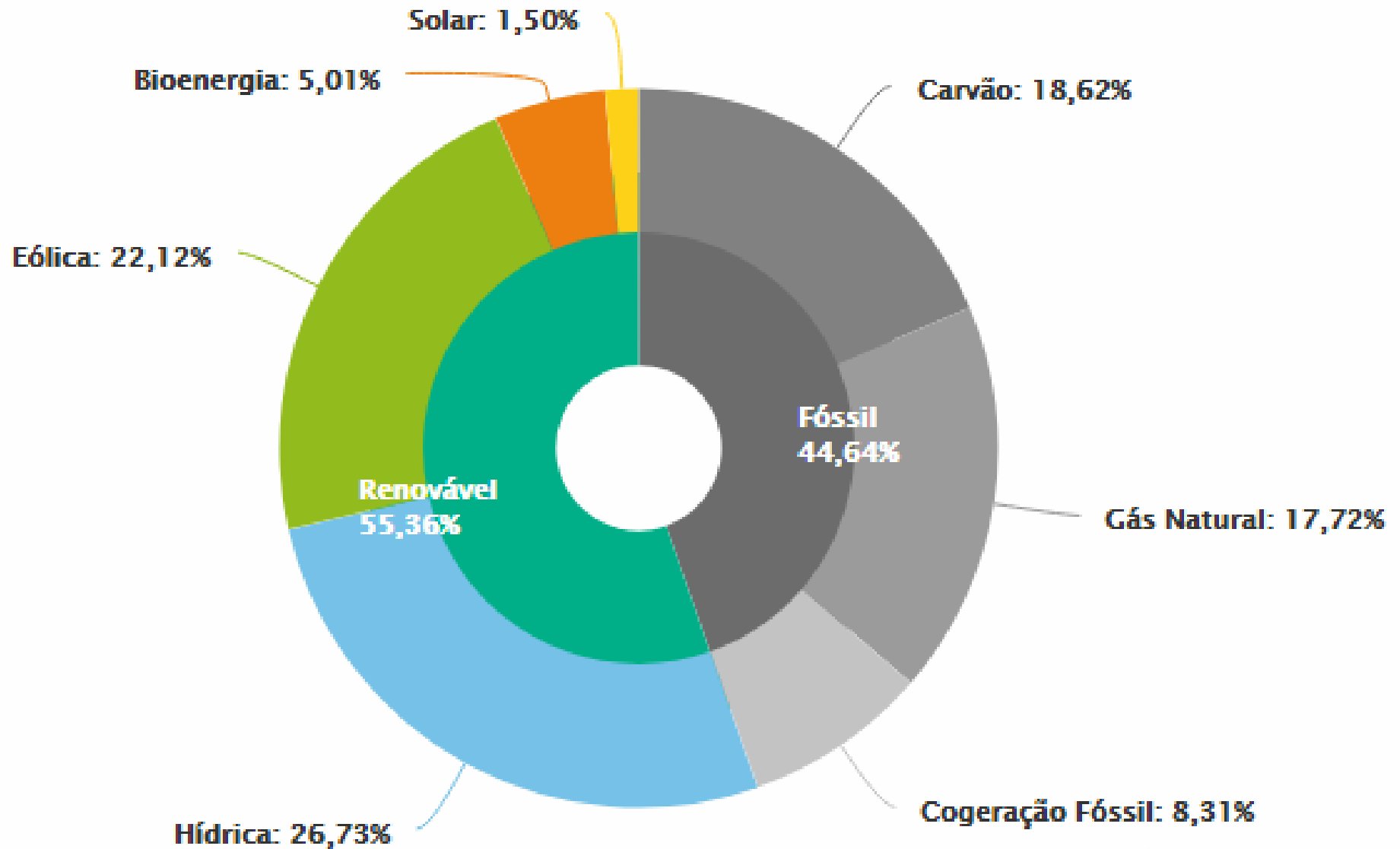


Share of energy from renewable sources in the EU Member States

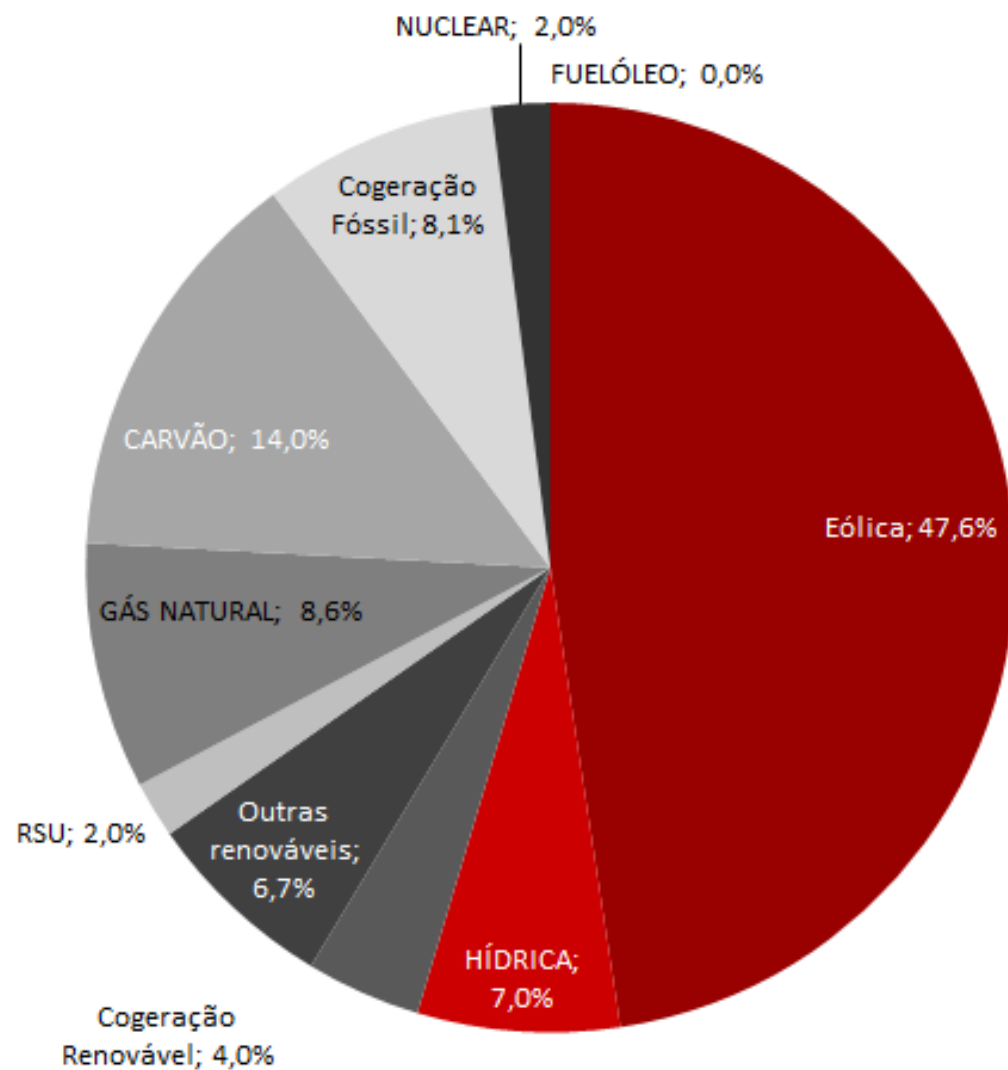
(in % of gross final energy consumption)



Produção de energia elétrica de janeiro a agosto de 2018



Repartição por tecnologia da energia comercializada pela EDP Serviço Universal



Fontes de Alimentação



- 63% **Termoelétrica**
- 19% **Hidroelétrica**
- 17% **Nuclear**
- 1% **Outras**

Fontes de Alimentação

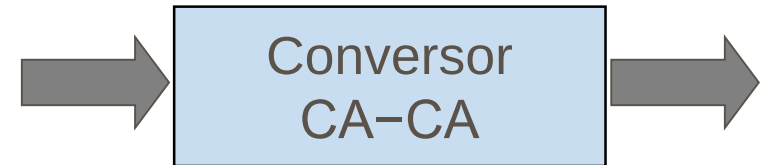
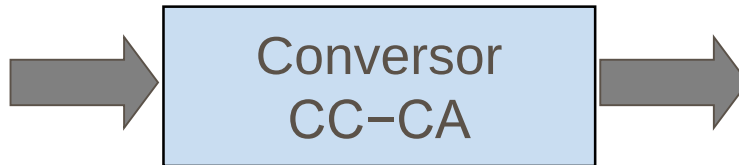
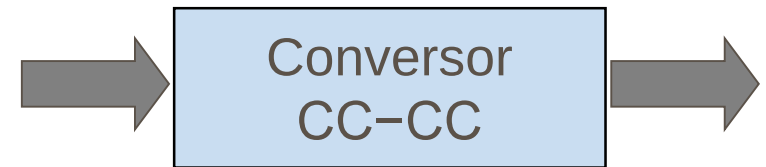
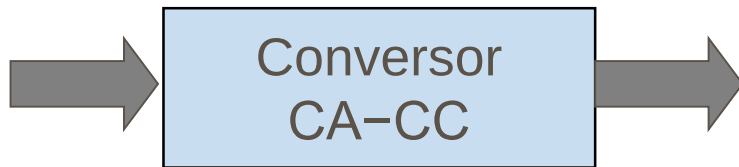


Fonte de alimentação de corrente contínua

Fontes de Alimentação



■ Outros conversores electrónicos de energia



Fontes de Alimentação

■ Diagrama de blocos

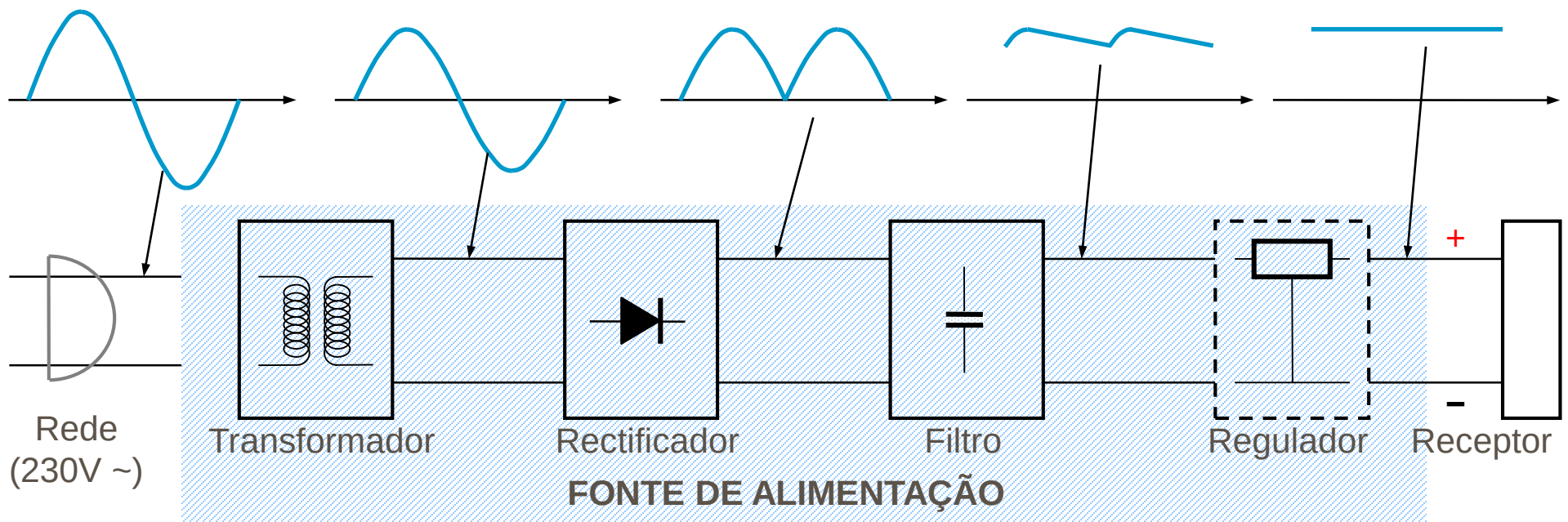
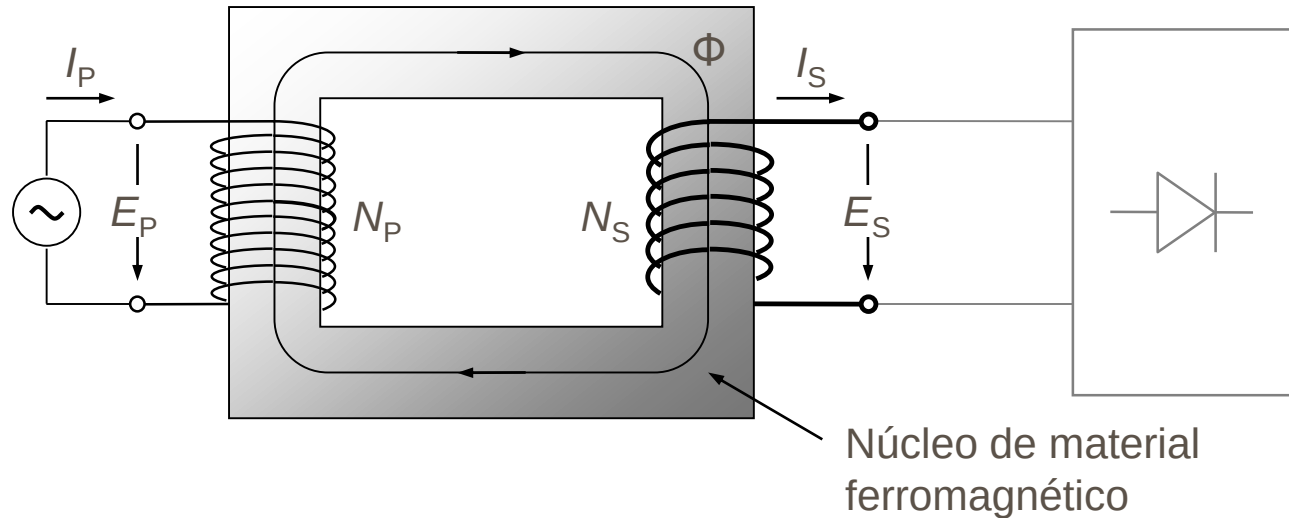


Diagrama de blocos duma fonte de alimentação de corrente contínua

■ Transformador



$$E_P = N_P \frac{d\Phi}{dt}, \quad E_S = N_S \frac{d\Phi}{dt} \quad \rightarrow \quad \frac{E_P}{E_S} = \frac{N_P}{N_S} = a$$

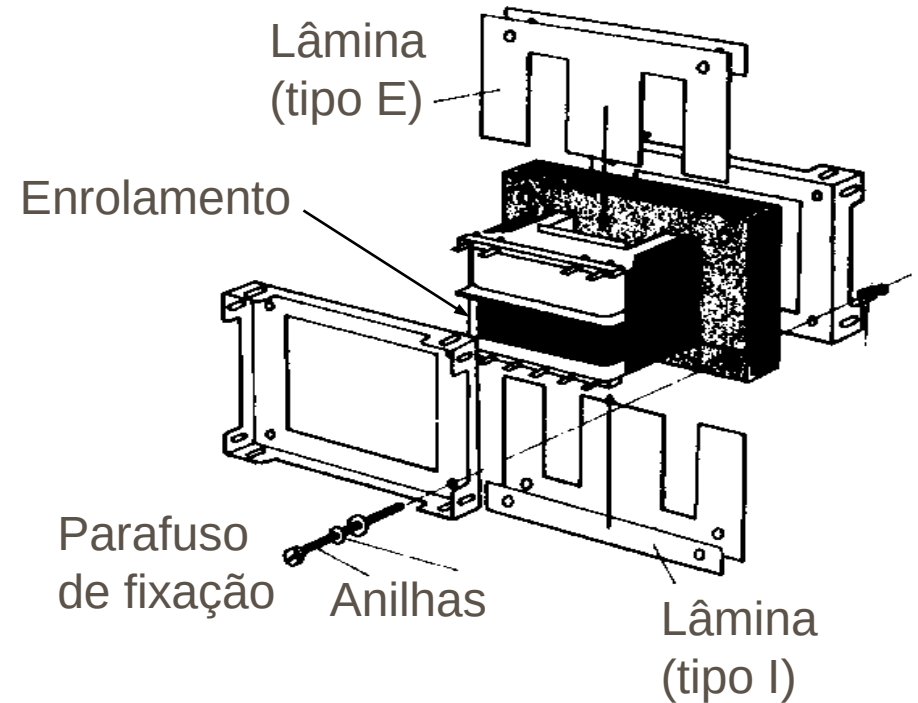
$$P_P = E_P \times I_P = E_S \times I_S = P_S \quad \rightarrow \quad \frac{I_P}{I_S} = \frac{N_S}{N_P} = \frac{1}{a}$$

Fontes de Alimentação

■ Transformador



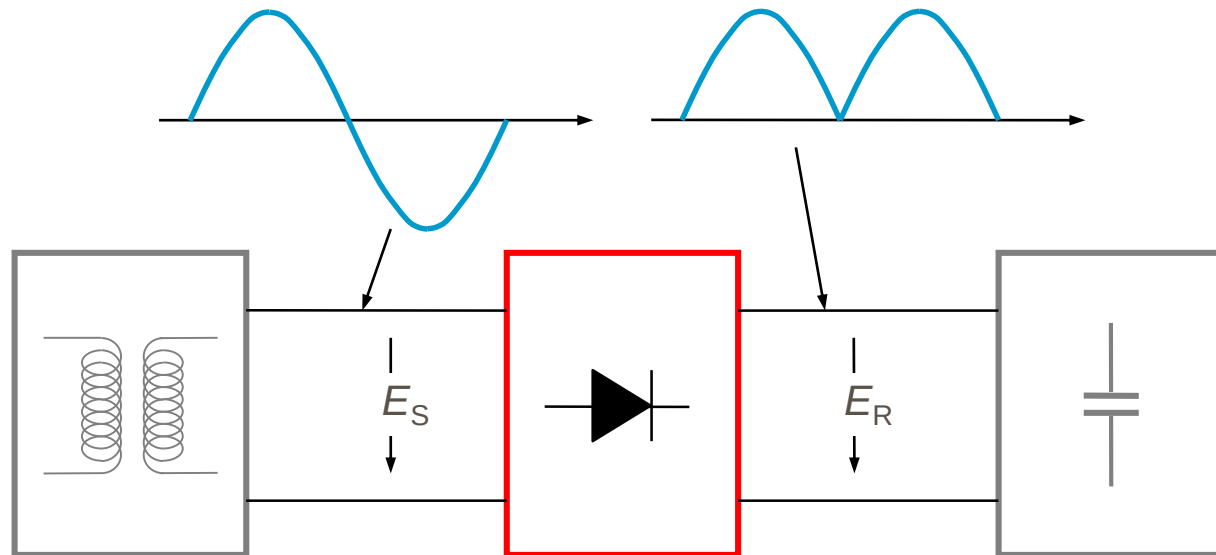
(a)



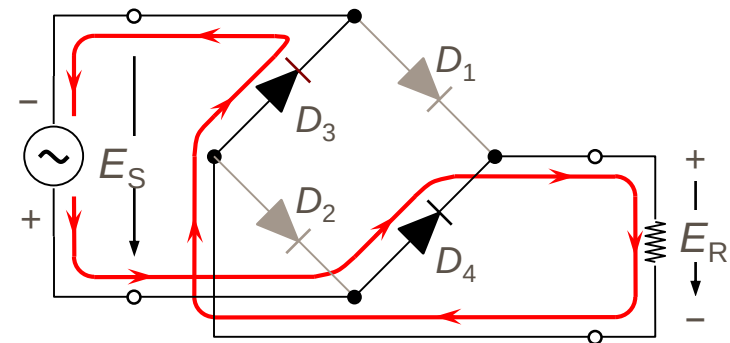
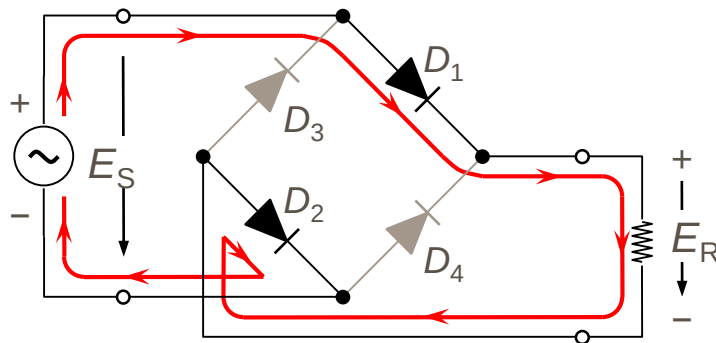
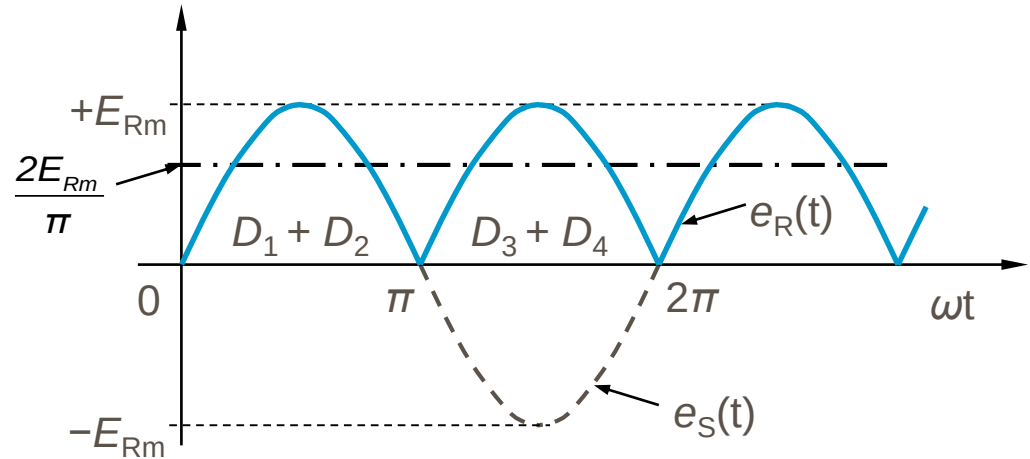
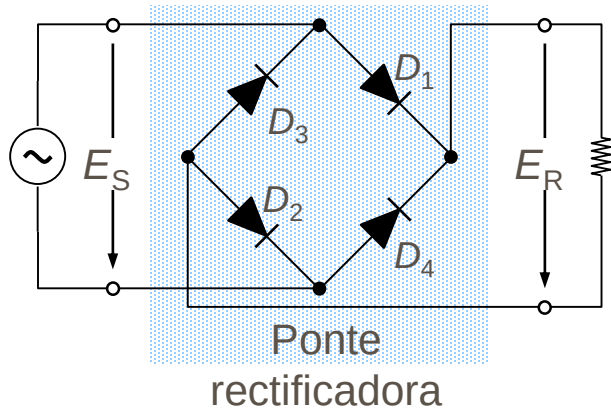
(b)

Transformador para fonte de alimentação:
(a) aspecto exterior; (b) construção

■ Rectificador



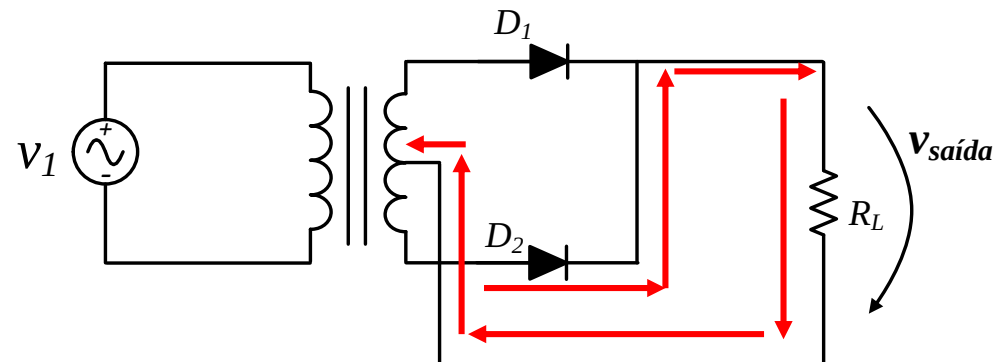
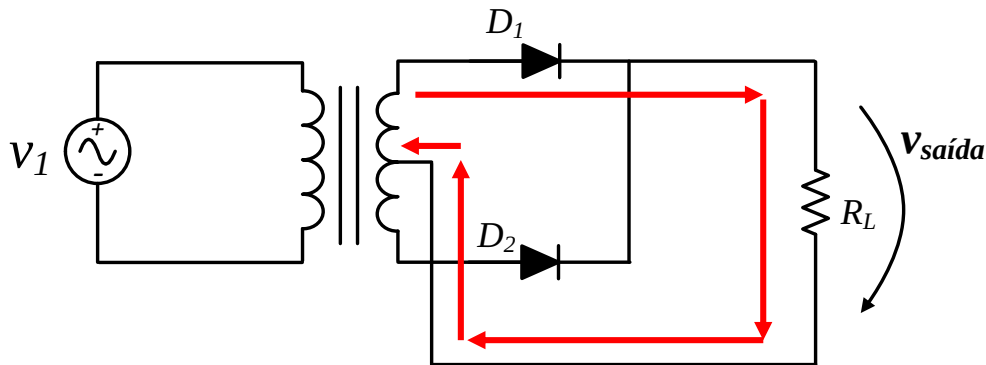
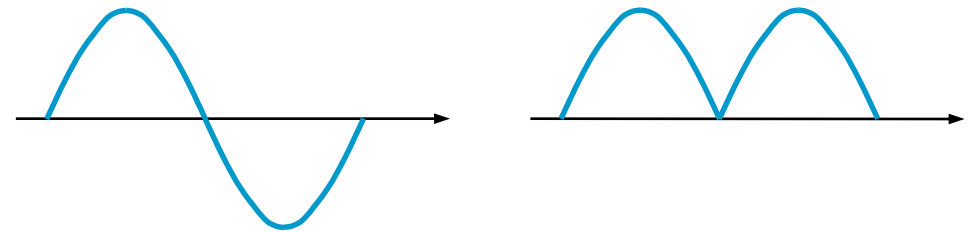
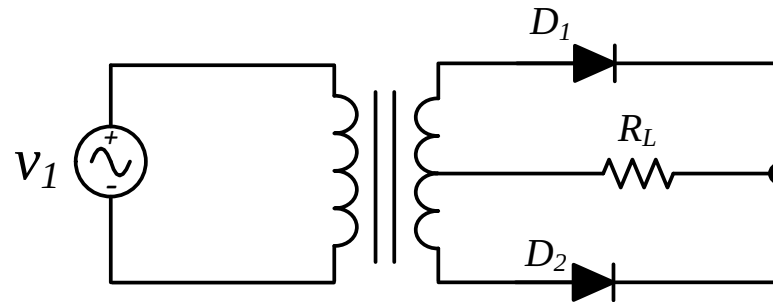
■ Rectificador



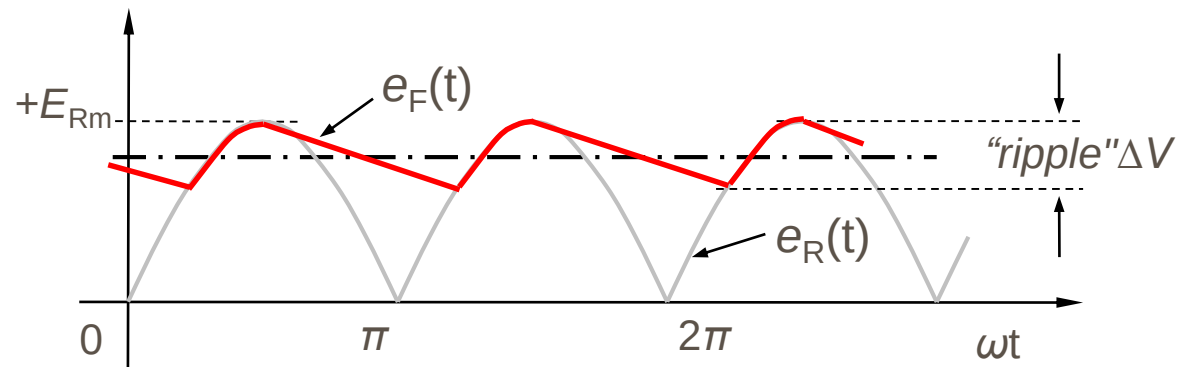
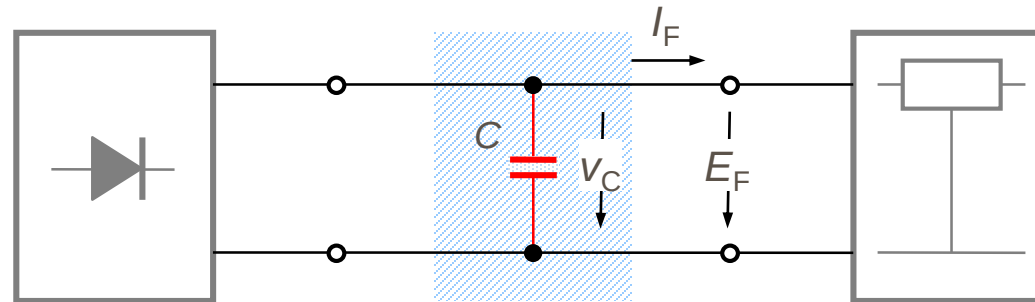
Funcionamento da ponte rectificadora

Fontes de Alimentação

■ Rectificador (transformador com ponto médio)



■ Filtro



$$i_C = C \frac{dv_C}{dt} = I_F \approx \text{cte} \rightarrow I_F \approx C \frac{\Delta V_C}{\Delta t} \rightarrow E_{Fr,p-p} = \Delta V_C = \frac{I_F}{C} \Delta t \rightarrow E_{Fr,p-p} \approx \frac{E_{Rm}}{f \times R_L C}$$

Fontes de Alimentação

■ Diagrama de blocos

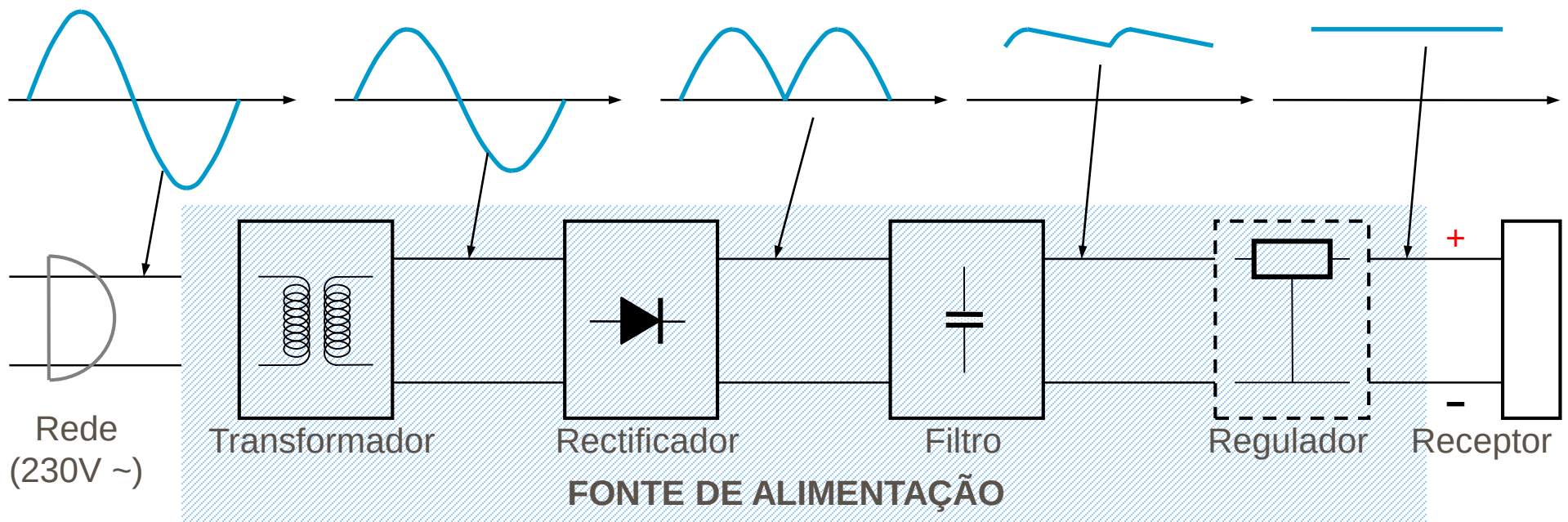


Diagrama de blocos duma fonte de alimentação de corrente contínua

Fim da Aula 3